

**GPU programming with CUDA** - GPUs and GPGPU, GPU architectures, Heterogeneous computing, Threads, blocks, and grids Nvidia compute capabilities and device architectures, Vector addition, Returning results from CUDA kernels, CUDA trapezoidal rule I, CUDA trapezoidal rule II: improving performance, CUDA trapezoidal rule III: blocks with more than one warp.

#### PRACTICAL COMPONENT OF IPCC

| Sl.NO | Experiments                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1     | Write a OpenMP program to sort an array on n elements using both sequential and parallel mergesort(using Section). Record the difference in execution time.                                                                                                                                                                                                                                                                                              |
| 2     | Write an OpenMP program that divides the Iterations into chunks containing 2 iterations, respectively (OMP_SCHEDULE=static,2). Its input should be the number of iterations, and its output should be which iterations of a parallelized for loop are executed by which thread.<br><br>For example, if there are two threads and four iterations, the output might be the following:<br>a. Thread 0 : Iterations 0 — 1<br>b. Thread 1 : Iterations 2 — 3 |
| 3     | Write a OpenMP program to calculate n Fibonacci numbers using tasks.                                                                                                                                                                                                                                                                                                                                                                                     |
| 4     | Write a OpenMP program to find the prime numbers from 1 to n employing parallel for directive. Record both serial and parallel execution times.                                                                                                                                                                                                                                                                                                          |
| 5     | Write a MPI Program to demonstration of MPI_Send and MPI_Recv.                                                                                                                                                                                                                                                                                                                                                                                           |
| 6     | Write a MPI program to demonstration of deadlock using point to point communication and avoidance of deadlock by altering the call sequence                                                                                                                                                                                                                                                                                                              |
| 7     | Write a MPI Program to demonstration of Broadcast operation.                                                                                                                                                                                                                                                                                                                                                                                             |
| 8     | Write a MPI Program demonstration of MPI_Scatter and MPI_Gather                                                                                                                                                                                                                                                                                                                                                                                          |
| 9     | Write a MPI Program to demonstration of MPI_Reduce and MPI_Allreduce (MPI_MAX, MPI_MIN, MPI_SUM, MPI_PROD)                                                                                                                                                                                                                                                                                                                                               |

#### Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

- Explain the need for parallel programming
- Demonstrate parallelism in MIMD system.
- Apply MPI library to parallelize the code to solve the given problem.
- Apply OpenMP pragma and directives to parallelize the code to solve the given problem
- Design a CUDA program for the given problem.

#### Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

**CIE for the theory component of the IPCC (maximum marks 50)**